Project 59

Interactive Social Guide

Decision making using a rule-based and an LLM-integrated navigation



Robot

Background

A guide robot must recognise interest in interaction while roaming the laboratory. Changing gaze and distance, and missed detections, make this decision difficult.

Objective

Develop a temporal social-state pipeline for Navel to support a comparison and evaluation of rule-based and LLM decisions in navigation scenarios

RQ1 How can gaze and distance observations over time be structured into a shared social-state representation for rule-based and LLM decision-making?

RQ2 How well can an LLM's decisions align with human preferences and appropriateness ratings, compared with those of a rule-based policy using the same representation?

Pipelin

Navel Sensors

Person detections, social cues and robot movement data

Track History

Person histories, changes in attention and distance, and observation quality

Shared Social State

A structured summary of the social context and available evidence

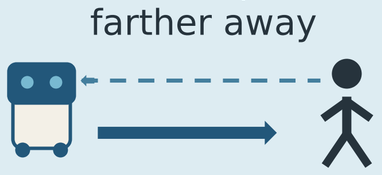
Decision Policies

Rule-based and LLM policies select an action from the same social state

Four possible actions

APPROACH

Interested person farther away



Move closer

ENGAGE

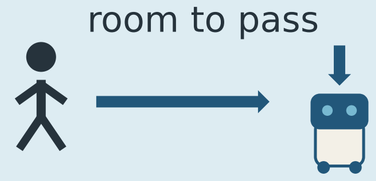
Interested person already nearby



Stop and greet

YIELD

Person needs room to pass



Give room to pass

CONTINUE

No sustained interest in interaction



Stay on the route

Result

Conclusion and future work

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